Final Year Project

Intelligent Voice Interview Platform

Complete Technical Documentation

March 2026 | Python 3.10/3.11 | Windows / Linux / macOS

Pipeline: Voice -> Whisper ASR -> Llama 3.2 LLM -> Score (duration-aware) + Behavioral Analysis

1. Features

1.1 Core Interview

- Automated voice interview with animated HR avatar
- Trilingual support: Arabic / French / English
- Smart follow-up questions when answer score < 8
- Per-question timer (configurable, default 90 s) with auto-stop
- Recording starts automatically after question audio ends
- Reconnection support - resumes at current question if in_progress
- Interview scheduling with 30-minute late access window

1.2 AI Pipeline

- ASR - faster-whisper (GPU-accelerated, ~1-3 s/answer)
- LLM evaluation - Ollama + Llama 3.2, strict grading 0-10, duration-aware scoring
- TTS - Edge-TTS 7.x (primary) + gTTS (automatic fallback)
- TTS prefetch - next question generated while candidate answers current one
- Facial analysis - HSEmotion EfficientNet-B0 (~82%) + MediaPipe FaceMesh

1.3 Privacy & Data

- All behavioral metrics stored in MongoDB - never sent to candidate
- Global HR report with per-answer facial summary + hiring decision
- Automatic recruiter notification on interview completion

1.4 Platform

- JWT authentication for recruiters
- PySide6 desktop client (Windows / Linux / macOS)
- MongoDB - candidates, positions, sessions, evaluations, notifications
- CSV / JSON export + analytics dashboard

2. Technical Pipeline

2.1 Main Flow

1. The candidate speaks -> PCM audio chunks sent via WebSocket
2. The backend assembles chunks into a WAV file
3. Whisper transcribes audio to multilingual text
4. MediaPipe FaceMesh + HSEmotion analyze facial behavior (server-side only)
5. Llama 3.2 evaluates the answer: score 0-10 enriched with facial metrics and response duration
6. Result signals sent to client (no scores or behavioral data exposed to candidate)
7. After all questions, global HR report generated with hiring decision

2.2 Pipeline Schema

```
Candidate speaks
  |  PCM chunks (WebSocket)
interview_handler.py  ->  audio buffer + video frames (2fps JPEG)
  |  parallel
  |-- asr_service.py (faster-whisper)  ->  transcript
  +-- facial_analysis_service.py (MediaPipe + HSEmotion)  ->  FacialMetrics
  |
llm_service.py  ->  evaluate_with_facial(
                      question, transcript,
                      facial_metrics,
                      duration_seconds, max_duration_seconds)
  |  score + verdict + needs_followup
interview_handler.py  ->  MongoDB (FacialAnalysisData embedded)
  |  WebSocket signal (no scores sent to candidate)
evaluation/service.py  ->  GlobalEvaluation + GlobalFacialSummary  ->  MongoDB
```

2.3 Duration-Aware Scoring

The LLM receives the actual response duration alongside the allowed maximum. Scoring is modulated by the usage ratio:

Usage ratio	Scoring impact
< 20% of allocated time	Penalty of -1 to -2 pts if content is also poor
40-90% of allocated time	Neutral - no impact (ideal range)
> 90% with rich content	Possible bonus of +0.5 pt
Short but precise and complete	No penalty applied

3. Prerequisites

Software	Version	Link	Notes
Python	3.10 or 3.11	python.org	Add to PATH during install
MongoDB	7.x Community	mongodb.com	Run as a service
Ollama	latest	ollama.com/download	Required for LLM evaluation
FFmpeg	6.0+	ffmpeg.org	Required only if gTTS fallback used
CUDA Toolkit	11.x or 12.x	developer.nvidia.com	Optional - GPU acceleration
VS C++ Build Tools	2022	visualstudio.com	Windows only - required for PyAudio

4. Installation

4.1 Clone the project

```
git clone https://github.com/ZeinebGhrab/sparkhire-ai.git
cd sparkhire-ai
```

4.2 Virtual environment

```
python -m venv .venv

# Windows PowerShell
.venv\Scripts\Activate.ps1
# Linux / macOS
source .venv/bin/activate
```

4.3 PyTorch (GPU recommended)

```
# CUDA 12.x
pip install torch --index-url https://download.pytorch.org/whl/cu121
# CPU only
pip install torch
```

4.4 Backend dependencies

```
pip install -r requirements.txt
```

4.5 Client dependencies

```
pip install -r client/requirements.txt
```

4.6 Facial analysis stack (install in this exact order)

timm==0.9.2 is required - timm==0.6.13 is missing timm.layers which causes a ModuleNotFoundError at model load.

```
pip install timm==0.9.2
pip install efficientnet_pytorch
pip install hsemotion
pip install mediapipe==0.10.14
pip install "protobuf>=4.25.3,<5.0.0"
```

Verify the install:

```
python -c "
from hsemotion.facial_emotions import HSEmotionRecognizer
fer = HSEmotionRecognizer(model_name='enet_b0_8_best_afew', device='cpu')
print('HSEmotion OK:', fer)
"
```

4.7 Environment file

```
cp .env.example .env
```

4.8 Whisper model

```
python scripts/download_whisper.py medium
```

Size	Weight	Use case
tiny	75 MB	Quick tests
base	145 MB	-
small	483 MB	-
medium (recommended)	**1.5 GB**	**Recommended**
large-v3	3.1 GB	GPU only - 6 GB+ VRAM

4.9 Pull LLM

```
ollama serve
ollama pull llama3.2
```

4.10 Seed database

```
python scripts/create_admin.py
python scripts/seed_job_positions.py
```

5. Running the Application

5.1 Backend

```
uvicorn backend.main:app --reload --host 0.0.0.0 --port 8000
```

Expected startup output:

```
  MediaPipe FaceMesh v5.2 | 478 landmarks + iris | CPU
  HSEmotion | model=enet_b0_8_best_afew | device=CPU
  Whisper 'medium' ready on CPU
  Edge-TTS initialized
  LLM Ollama available | model=llama3.2
```

API documentation (Swagger): http://localhost:8000/docs

Health check: http://localhost:8000/health

5.2 Desktop Client

```
python -m client.main
```

The client displays a language selection screen, then asks for the session_id.

6. Project Architecture

```
sparkhire-ai/
|-- backend/
|   |-- main.py
|   |-- config.py
|   |-- database.py
|   |-- auth/                        <- JWT authentication
|   |-- candidates/                  <- Candidate CRUD
|   |-- interviews/
|   |   |-- crud.py                  <- completion notification trigger
|   |   |-- models.py                <- Question.weight, FacialAnalysisData
|   |   +-- routes.py
|   |-- evaluation/
|   |   |-- models.py                <- GlobalEvaluation, FacialSummary
|   |   |-- service.py               <- weighted avg + facial injection
|   |   +-- routes.py
|   |-- services/
|   |   |-- asr_service.py           <- Whisper GPU/CPU
|   |   |-- tts_service.py           <- Edge-TTS 7.x + gTTS fallback
|   |   |-- edge_tts_engine.py       <- retry x3, 7.x API
|   |   |-- llm_service.py           <- evaluate_with_facial() + duration
|   |   |-- facial_analysis_service.py  <- v5.2: MediaPipe + HSEmotion
|   |   +-- avatar_service.py
|   |-- websocket/
|   |   |-- connection_manager.py
|   |   +-- interview_handler.py    <- video_frame, metrics backend-only
|   |-- notifications/
|   |-- analytics/
|   +-- export/
|
|-- client/
|   |-- main.py
|   |-- config.py
|   |-- core/
|   |   |-- audio_recorder.py
|   |   |-- video_recorder.py       <- 2fps JPEG frame collector
|   |   +-- websocket_client.py
|   +-- ui/
|       |-- main_window.py           <- session flow, no metric display
|       |-- interview_widget.py      <- progress + recording controls only
|       |-- video_player_widget.py
|       |-- camera_preview_widget.py <- PiP camera + REC badge only
|       +-- stark_theme.py
|
|-- docs/                            <- architecture diagrams (PNG)
|-- scripts/
|-- models/
|-- uploads/
|-- assets/videos/
|-- requirements.txt
|-- client/requirements.txt
+-- .env.example
```

7. Facial Behavior Analysis

7.1 Detection pipeline

```
Webcam frames (2fps, JPEG)
  |
  |-- MediaPipe FaceMesh
  |     478 landmarks + iris (468-477)
  |     * EAR          -> blink detection (threshold 0.22)
  |     * Iris offset -> eye contact
  |     * solvePnP     -> yaw / pitch / roll (normalized v5.1)
  |     * Lip corners -> smile
  |     * Brow raise / frown -> Action Units AU1/2/4
  |
  +-- Emotion backend (priority order)
        1. HSEmotion EfficientNet-B0  ~82%  AffectNet8+AFEW  <- active
        2. DeepFace VGG CNN            ~73%  AffectNet        <- fallback 1
        3. FACS heuristics             ~60%  landmarks only   <- fallback 2
```

7.2 Metrics stored per answer (MongoDB)

Metric	Range	Description
`confidence_score`	0-10	Eye contact + stability + positive emotions - negative emotions
`stress_score`	0-10	Brow frown + instability + fear + angry + sad
`engagement_score`	0-10	Eye contact + expressiveness + smile - sad - disgust
`eye_contact_ratio`	0-1	Fraction of frames with gaze directed at camera
`head_stability`	0-1	1 - (yaw_std + pitch_std) / 60
`smile_ratio`	0-1	Fraction of frames with smile detected
`blink_rate`	bpm	Estimated at 10fps; returns 0.0 if < 5 frames
`dominant_emotion`	string	Most frequent emotion across all frames

7.3 formula

Issue in v5.1	Fix in v5.2
eye_contact dominated confidence (4.0 pts) - staring blankly scored high	Reduced to 2.5 pts; positive emotions contribute 2.0 pts
sad not penalizing confidence or engagement	sad reduces confidence (-0.8x) and engagement (-1.5x)
stress_score ignored sad and surprise	Both now contribute to stress (+1.0 and +0.5)
blink_rate assumed 2fps -> 0.0 bpm on short answers	Corrected to 10fps; returns 0.0 if < 5 frames

7.4 Privacy - behavioral data flow

```
Backend analysis -> MongoDB (evaluations collection)
                         |
                  HR report only
                  (never sent to candidate client)
```

The candidate client receives only:

- answer_evaluated -> { question_order, had_followup, is_initial }
- global_evaluation -> { decision, decision_label, decision_color, totals }

8. Scoring & Hiring Decisions

8.1 Weighted average

Each question has a configurable weight (default 1.0, range 0.1-10.0):

```
average_score = sum(score_i x weight_i) / sum(weight_i)
```

Question	Score	Weight	Points
Q1 - Presentation	8	1.0	8
Q2 - ML tools	6	2.0	12
Q3 - Missing data	7	3.0	21
Weighted result	**6.83**	**6.0**	**41 / 6 = 6.83**

8.2 Grading scale

Score	Verdict	Meaning
9-10	Excellent	Precise, concrete, exemplary - rare technical mastery
7-8	Very Good	Good but lacking depth or specific examples
5-6	Acceptable	Superficial or vague, notable inaccuracies
3-4	Poor	Errors or partial understanding
0-2	Insufficient	Off-topic, incorrect, or empty

Score < 8 automatically triggers a follow-up question.

8.3 Hiring decision thresholds

Average /10	Decision	Color code
>= 7.0	Accepted	#10B981 (green)
5.0 - 6.9	On Hold	#F59E0B (amber)
< 5.0	Rejected	#EF4444 (red)

9. WebSocket Message Contract

9.1 Server -> Client (candidate-facing only)

Type	Payload fields	Notes
`welcome`	total_questions, position_title, candidate_name, expires_at, language	Session start
`welcome_back`	same + current_question_index, is_reconnection	On reconnection
`question`	order, weight, max_duration, progress, language	Question audio
`answer_saved`	duration, question_order, saved	Acknowledge receipt
`answer_evaluated`	question_order, had_followup, is_initial	No scores or feedback
`answer_followup_completed`	question_order, had_followup	No scores or feedback
`global_evaluation`	decision, decision_label, decision_color, totals	No average score or summary
`interview_completed`	total_questions, total_answers, position_title, candidate_name	Session end
`error`	message, error_type	Error signal

9.2 Client -> Server

Type	Payload	Description
`audio_chunk`	audio_data: base64 PCM	Audio during recording
`video_frame`	data.frame: base64 JPEG	Camera frame for facial analysis
`answer_complete`	-	Recording stopped
`audio_finished`	-	Client finished playing audio
`end_interview`	-	Candidate ended early

10. API Reference

10.1 Authentication

Method	Route	Description
POST	`/auth/login`	Recruiter login -> JWT token
GET	`/auth/me`	Connected recruiter profile

10.2 Interviews

Method	Route	Description
POST	`/interviews/sessions`	Create a session
GET	`/interviews/sessions`	List all sessions
GET	`/interviews/sessions/{id}`	Get session details
WS	`/ws/interview/{session_id}?lang=fr`	WebSocket interview endpoint
POST	`/interviews/positions`	Create a job position with questions
GET	`/interviews/positions`	List all job positions

10.3 Evaluations

Method	Route	Description
POST	`/evaluations/`	Trigger evaluation for a session
GET	`/evaluations/{session_id}`	Get HR report (includes facial data)

10.4 Notifications

Method	Route	Description
GET	`/notifications/`	List all notifications
GET	`/notifications/unread-count`	Unread badge count
PATCH	`/notifications/{id}`	Mark as read
POST	`/notifications/mark-all-read`	Mark all as read
DELETE	`/notifications/{id}`	Delete a notification

10.5 Analytics & Export

Method	Route	Description
GET	`/analytics/dashboard`	All KPIs combined
GET	`/analytics/scores`	Accepted/Rejected KPIs + 6-month trend
GET	`/analytics/system`	System health + storage
GET	`/export/candidates/csv`	All candidates CSV

GET	`/export/interviews/csv`	All sessions CSV
GET	`/export/interviews/{id}/json`	Full interview report JSON (HR only)

11. Configuration (.env)

Variable	Default	Description
MONGODB_URL	mongodb://localhost:27017	MongoDB connection URL
MONGODB_DB_NAME	sparkhire_ai	Database name
SECRET_KEY	change-me	JWT key - generate with secrets.token_hex(32)
WHISPER_DEVICE	cuda	cpu or cuda (NVIDIA GPU)
WHISPER_COMPUTE_TYPE	float16	float16 (GPU) or int8 (CPU)
OLLAMA_URL	http://localhost:11434	Ollama server URL
OLLAMA_MODEL	llama3.2	LLM model: llama3.2, mistral, etc.
FACIAL_ANALYSIS_ENABLED	true	Enable/disable facial analysis
FACIAL_CAPTURE_FPS	2.0	Frames per second sent from client
WEBSOCKET_URL	ws://localhost:8000	WebSocket URL used by client
KMP_DUPLICATE_LIB_OK	TRUE	Windows only - OpenMP conflict fix

12. MongoDB Schema

12.1 Collections

- recruiters - email, password_hash (bcrypt)
- candidates - first_name, last_name, contact, technical_skills, experiences, education, embeddings
- job_positions - title, department, questions[] (order, question_ar/fr/en, weight, max_duration_seconds)
- interview_sessions - session_id, candidate_id, job_position_id, status, language, answers[], evaluation_score, evaluation_decision
- evaluations - session_id, per_answer[].facial, facial_summary, average_score, decision, key_strengths, key_improvements
- notifications - recipient_email, type, title, message, read, priority

12.2 answers[] embedded in interview_sessions

```
answers: [
  {
    question_order: 1,
    transcript: '...',
    duration_seconds: 45.2,
    evaluation: {
      score: 7.5,  verdict: 'Very Good',  weight: 2.0,
      had_followup: false,  llm_model: 'llama3.2',
      facial_analysis: {
        dominant_emotion: 'neutral',
        emotion_scores: { happy: 10.1, neutral: 74.3, sad: 8.2 },
        eye_contact_ratio: 0.87,  confidence_score: 6.9,
        stress_score: 4.6,  engagement_score: 6.8,
        frames_analyzed: 25,  frames_with_face: 24
      }
    }
  }
]
```

12.3 Global facial summary in evaluations

```
facial_summary: {
  avg_confidence: 6.2,  avg_stress: 4.4,  avg_engagement: 6.0,
  avg_eye_contact: 0.72,  avg_head_stability: 0.75,
  dominant_emotion: 'neutral',  facial_available: true
}
```

13. Tech Stack

Layer	Technology	Version	Role
Backend	FastAPI + Uvicorn	0.115 / 0.32	REST API + WebSocket
Database	MongoDB + Motor	7.x / 3.5	Data persistence
ASR	faster-whisper + ctranslate2	1.0.3 / 4.4.0	GPU transcription
TTS primary	Edge-TTS	7.2.7	Microsoft neural TTS - fixed 403
TTS fallback	gTTS	2.5.3	Google TTS automatic fallback
LLM	Ollama + Llama 3.2	latest	Answer evaluation + duration scoring
Landmarks	MediaPipe	0.10.14	478 landmarks + iris gaze
Emotions	HSEmotion EfficientNet-B0	latest	~82% accuracy on AffectNet8
Emotions fallback	DeepFace VGG CNN	0.0.99	~73% accuracy on AffectNet
timm	timm	0.9.2	Required by HSEmotion
Auth	python-jose + bcrypt	3.3 / 4.2	JWT + password hashing
Client GUI	PySide6	6.7	Qt framework
Audio	pygame + PyAudio + pydub	2.6 / 0.2 / 0.25	Playback + capture + conversion
Video	OpenCV	4.10	Frame capture + decode

14. Performance

Measured on RTX 4050 Laptop (6.4 GB VRAM):

Component	CPU	GPU
Whisper ASR (medium)	5-10 s	~1-3 s
Llama 3.2 evaluation	15-30 s	~2-8 s
Edge-TTS synthesis	~1 s	~1 s (network)
HSEmotion per frame	~40 ms	~8 ms
MediaPipe per frame	~8 ms	~8 ms (CPU only)
Total per question	**25-45 s**	**~5-12 s**

15. Troubleshooting

Error	Cause	Fix
`No module named 'timm.layers'`	timm 0.6.13 too old	`pip install timm==0.9.2`
`No module named 'efficientnet_pytorch'`	Missing HSEmotion dep	`pip install efficientnet_pytorch`
`Aucun modele HSEmotion disponible`	Stack not installed	`pip install timm==0.9.2 efficientnet_pytorch hsemotion`
`FieldDescriptor has no attribute label`	protobuf >= 5 conflict	`pip install "protobuf>=4.25.3,<5.0.0"`
`cudnn_ops_infer64_8.dll not found`	cuDNN 8 missing	nvidia-cudnn-cu12==8.9.7.29 in requirements
`float16 not supported`	CPU + float16	`Set WHISPER_COMPUTE_TYPE=int8`
`OMP: Error #15`	OpenMP conflict	`Set KMP_DUPLICATE_LIB_OK=TRUE in .env`
`Edge-TTS 403 Forbidden`	edge-tts 6.x old token	`pip install edge-tts==7.2.7`
`Facial analysis timeout`	MediaPipe failing	Fix protobuf version, restart server
`PyAudio not found`	PortAudio missing	`pipwin install pyaudio (Windows)`
`Connection refused :11434`	Ollama stopped	`Run: ollama serve`
`MongoDB timeout`	Service stopped	`net start MongoDB (Windows)`
`Empty transcription`	Audio too short	Speak clearly for at least 1 second

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